

MANISH HIRA

Data Science in Marketing & Finance

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SUMMARY-----

- ◆ Demonstrated leadership across financial services, consulting, and technology, with deep technical proficiency and a record of driving data-driven strategic growth.
- ◆ Proficient in multi-platform AI tools, cross-functional management, and publishing impactful research/patents

EXPERIENCE-----

CAPITAL ONE | DATA ANALYSIS MANAGER | Richmond, VA

Jan 2019 – present

- ◆ Led several analytical projects for brand and marketing to develop meaningful insights and inform strategic decision making for the brand organization
 - ◆ Evaluated advertising experiments for Venture X and Savor product launch in 10 markets and studied the impact of local advertising in select channels overlaid on top of national campaign. Informing \$50 MM media allocation
 - ◆ Impacted the executive leadership decision to allocated a baseline brand marketing spend of 20% of the total marketing budget, using a collection of media investment analysis over last 20 years which included studying natural experiments related to product launches.
 - ◆ Shipped Tableau executive dashboards cutting time to insight for marketing partners by 40-50%
 - ◆ Introduced framework to centralize data from multiple sources and democratized its access for informed decision making and help answer strategic questions for the leadership
 - ◆ Analyzed metrics across dimensions such as geography, channel, competitive media spend, website traffic, share of direct mail volumes, competitive credit card spend etc.
 - ◆ Proposed strategic recommendations to the business to drive roughly 20% higher growth in the select product category and test markets
- ◆ Drafted a long term strategic vision for the analytics stack incorporating data sources and agentic analytical capabilities utilizing the Meta's LLAMA model in a GenAI sandbox, with potentially 30% operational efficiency gains
- ◆ Engineered a predictive propensity model to capture wallet share, using XGBoost to identify high-value market opportunities, by integrating Experian TAPS universe credit spending data. Informed data -driven recommendations for incremental strategic growth in spend of 5-10% in select geographies
- ◆ Knowledge and thought leadership - attend conferences like AudienceXScience to gather information on ongoing trends and share with broader department
 - ◆ Conceptualized and developed a "State of the Business" newsletter with insights on business performance, state of brand and the industry for executive audience
 - ◆ Explored and evaluated several analytics tools for data aggregation, analysis and presentation
- ◆ Product managed for end to end marketing analytics stack of the Capital One's brand organization
- ◆ Leadership and management
 - ◆ Managed a team of analysts, data engineers as well as product backlog, data and analytical needs
 - ◆ Extensive process of Talent management and performance management against the department and job family benchmark behaviors
 - ◆ Communicate insights to broader teams and manage both up and down the hierarchy
 - ◆ Attended Capital One Leads training program
 - ◆ Technical and data management for governance, compliance
- ◆ Utilized Neustar's DSDK (data science kit) to analyze use cases such as optimum conversion frequency, cross-lob overlap and delivery against target audiences and leading to cost optimization recommendations
- ◆ Utilized advanced analytics platforms such as Databricks in AWS cloud to work with large data sets and presented the results on easily digestible dashboards and decks using Tableau as the front end
- ◆ Built multiple quantitative solutions and dashboards to inform views on competitive landscape, brand recognition and try to solve core business problems in the AdTech, MarTech space
- ◆ Attended PADP (professional analyst development program) a signature program for the analyst community

STATE OF VIRGINIA, CAPITAL ONE | DATA ANALYTICS CONSULTANT Nov 2015 – Dec 2018

- ◆ Data and analytics consulting assignments for financial services and retail clients
- ◆ Delivered targeted card marketing campaigns, performed segmentation and targeting of product offerings to more than 50 million customers, and \$500 million quarterly transactions
- ◆ Worked on customer management campaigns reaching a customer base of more than 60 million accounts
- ◆ Built machine learning models using credit card transaction data to predict first time shoppers at retail partners using algorithm xgboost with an AUC score of 0.81.
- ◆ Translating business requirements into dimensional data models and maintaining data element definitions
- ◆ Design and develop cloud based ETL/ELT pipelines for retail operations of a state retail agency
- ◆ Data engineering from source to staging to enterprise data warehouse using Unix scripting and Oracle

SAMSUNG ELECTRONICS | SENIOR MANAGER | South Korea & India Jan 2009 – Aug 2013

- ◆ Planned & Managed a product portfolio of Samsung mobile products impacting sales of about 100 million units
- ◆ Founded the Intellectual Property Cell for Innovation Management generating *18 granted patents* and institutional IP pipeline
- ◆ Developed Business Intelligence reporting dashboards using MS SQL Server, SSIS/SSRS to extract and present insights from operational data sources
- ◆ Demonstrated ability to drive project life cycle, supported strategic planning to execution for multiple projects and balance priorities
- ◆ Drove solutions across development, QA, marketing and design teams to resolve issues, manage product management process for project deliverables
- ◆ Mentored a team of engineers to develop embedded software for low-end ultra-low cost mobile devices, which are constrained on compute and memory resources

WIPRO TECHNOLOGIES | TECHNICAL LEAD | USA, Finland, Japan, India Sep 2000 – Jan 2009

- ◆ Developed multimedia projects worth more than \$10 million for Qualcomm, resolving OEM (*HTC, Motorola and Samsung*) commercialization issues for *MSM7x00 chips*
- ◆ Implemented automated mobile test framework for *Nokia Mobile* in Finland. Tested the Linux based open source software system on Texas Instruments' OMAP platform
- ◆ Coordinated development of MRI scanning software according to DICOM standards *worth \$2 million for Toshiba*

EDUCATION-----

VIRGINIA COMMONWEALTH UNIVERSITY

PhD Candidate *SYSTEMS MODELING & ANALYSIS* 2024-till date *Coursework* - Mathematical Statistics, Linear Modeling, Research Exploration, Graph Theory, Optimization

The Ph.D. in Systems Modeling and Analysis is offered jointly by the Department of Statistical Sciences and Operations Research and the Department of Mathematics and Applied Mathematics. The program focuses on the development of the mathematical and computational skills used to conceptualize and analyze real-world systems

Working paper - sufficient dimension reduction; aligning research to causal inference and high-dimensional modeling relevant to marketing analytics/credit risk.

UNIVERSITY OF WASHINGTON

MS COMPUTATIONAL FINANCE & RISK MANAGEMENT 2019-2021 *Coursework* - Investment Science, Financial Data Science, Asset Allocation & Portfolio Management, Options and Derivatives, Monte Carlo methods in Finance, Optimization Methods in Finance, Risk in Financial Institutions, Credit Risk Management, Machine Learning for Finance, Partial Differential Equations (GPA: 3.7)

Projects -

- Mortgage Default Credit Risk Model using Machine Learning - Used Machine Learning Algorithms to detect the top loan defaulters post the 2007 financial crisis using Fannie Mae data set. The classification algorithms used were Decision tree, Random Forest XG Boost, Logistic Regression, KNN, DNN and CNN. Recognized as one of the best projects in class by the Professor.

- Retirement Portfolio with Market Forecast - Implemented a recursive dynamic programming model to design and manage retirement portfolios for 6 client profiles with prior savings and specific constraints; recommended the best possible asset allocations for each of these clients on the basis of market forecasts and the results from the constrained recursive problem.

UNIVERSITY OF NOTRE DAME

MBA CONSULTING & FINANCE (*Graduated Cum Laude*) 2013 – 2015 *Arthur Foundation Notre Dame MBA Fellow* (100% tuition award) || GMAT: 750 || GPA: 3.6 *Finance & Analytics Coursework* – Statistical Inference, Forecasting & Predictive Analytics, Market Research, Corporate Finance, Economics, Investments, Equity Analysis, Fixed Income Process, Mergers and Acquisitions, Global Investment Management

UNIVERSITY OF DELHI, NSIT

BACHELOR of ENGINEERING (*INSTRUMENTATION & CONTROL*) 1996 – 2000 Consultant (intern) for *Center for Development of Telematics* (Government funded R&D project, a Linux based networking application using socket programming)

CERTIFICATIONS-----

- ♦ *AWS Machine Learning - Speciality*
- ♦ *AWS Data Analytics - Speciality* - DynamoDB, Kinesis Analytics, Amazon Elasticsearch Service, Redshift, RDS, Aurora, Lambda, Glue ETL, Quicksight
- ♦ *MLOps Specialization (Coursera)* - machine learning in production, machine learning data lifecycle in production, machine learning modeling pipelines, deploying machine learning models in production
- ♦ *Machine Learning Engineer Nanodegree (Udacity)* projects - Supervised Learning, Unsupervised Learning, Reinforcement Learning, Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks); unsupervised learning (clustering, dimensionality reduction, recommender systems, deep learning), anomaly detection, regularization, dimensionality reduction
- ♦ *Machine Learning (Coursera Stanford)* - supervised, unsupervised learning, SVM, NNs
- ♦ *Quantitative Analysis in R (Datacamp)* - Time series analysis, ARIMA, Forecasting, Portfolio analysis, Credit Risk, Bond Valuation, Trading in R

SKILLS-----

Programming Language: Python, SQL, PySpark, R, C/C++, Unix Shell, PyTorch, TensorFlow

Tools: Snowflake, DataBricks, Tableau, Looker, PowerBI, Airflow, ODI, Teradata SQL Platforms:

Amazon Web Services (AWS), EMR, Glue, Lambda, Sagemaker, Jupyter, Linux

Project Management, Configuration Management: Agile (JIRA), Git/Github, Perforce, CVS

Algorithms / Libraries: Window function, Query performance tuning, Machine Learning, Linear/Logistic/Lasso Regression, Decision Trees, Random Forest, Support Vector Machines, Clustering, probabilistic models (Gaussian, Bayesian neural network), A/B testing

ML Libraries: Scikit-Learn, Pandas, Matplotlib, Seaborn, NumPy, SciPy, ggplot2, pySpark, Plotly, QuantLib, xgboost

PUBLICATIONS-----

- ♦ Google Scholar Profile - <https://scholar.google.com/citations?user=XBCYoOUAAAAJ>
- ♦ Patent for Capital One - Smart Card Authentication - US11836733B2
- ♦ Patent for Samsung Electronics Near Field Communication and apparatus thereof - US9160418B2

VOLUNTEERINGS-----

- ♦ Provided data and technology consulting services to local small businesses through engagements with Taproot Foundation, PHA, and Startup Virginia.
- ♦ Served as a judge at CES Awards, Codie Awards 2025 in the AI enabled education solutions category, AI for Good Hackathon '25 and at VCU Graduate school 28th annual research symposium
- ♦ IEEE Senior Member 101554119